AP PRECALCULUS

Unit 1 • Worksheet 1.2D • Answer key

Worksheet 1.2D • Rates of Change

Answer key with worked solutions

Directions

Answer every question. Multiple-choice questions have one best answer. On free-response questions, show your work and justify your answers. Questions labeled **NO CALCULATOR** must be done without a calculator; questions labeled **CALCULATOR** require a graphing calculator. For questions with a graph, read the values you need from the grid.

Question 1

NO CALCULATOR

Using the interval $[2,\ 2.001]$, which is the best estimate of the rate of change of $f(x) = x^2 + x$ at $x = 2$?

(A) 4

(B) 6

(C) 2.5

(D) 5

Solution

Answer: (D). $\frac{f(2.001)-f(2)}{0.001} = \frac{6.005001-6}{0.001} = 5.001$.

Question 2

NO CALCULATOR

The function g changes from increasing to decreasing at $x = 3$. Which must be true about the rate of change of g at $x = 3$?

(A) It changes from negative to positive.

(B) It changes from positive to negative.

(C) It is undefined.

(D) It is the greatest value of g.

Solution

Answer: (B). Increasing means positive rate of change; decreasing means negative.

Question 3

CALCULATOR

Which is the best estimate of the rate of change of $f(x) = \sqrt{x}$ at $x = 4$?

(A) 0.125

(B) 0.25

(C) 0.5

(D) 2

Solution

Answer: (B). $\frac{\sqrt{4.001}-2}{0.001} \approx 0.24998$, and from the left ≈ 0.25002: about 0.25.

Question 4 (Free response) NO CALCULATOR

A ball is thrown upward. Its height is $h(t) = -5t^2 + 20t + 2$ meters, t seconds after it is thrown.

(a) Find the average velocity of the ball on $[1, 2]$, $[1, 1.1]$ and $[1, 1.01]$.
(b) Use (a) to estimate the velocity of the ball at $t = 1$, with units.
(c) Estimate the velocity at $t = 3$ using $[2.99, 3]$ and $[3, 3.01]$. What does its sign mean?

Solution
(a) $\frac{h(1+k)-h(1)}{k} = 10 - 5k$: 5, 9.5, 9.95 m/s.
(b) About 10 m/s.
(c) $\frac{h(3+k)-h(3)}{k} = -10 - 5k$: $[2.99, 3]$ ($k = -0.01$): -9.95; $[3, 3.01]$: -10.05. About -10 m/s: the ball is falling.

Question 5 (Free response) NO CALCULATOR

Explain, using rates of change, why the graph of a function that is increasing and concave down rises more and more slowly.

Solution
Increasing means the rate of change is positive. Concave down means the rate of change is decreasing. A positive rate of change that keeps decreasing makes the outputs increase by smaller and smaller amounts over equal intervals.